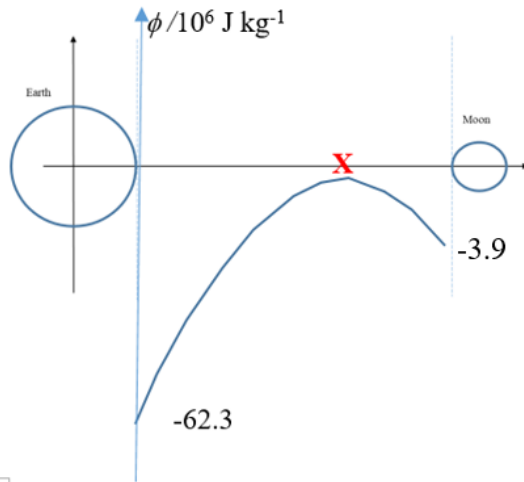


2743 ~ 2.54 0050 W.

Q3(a)



3(b)(i)

The neutral point is the position along a straight line between the Earth and the Moon where an object experiences no net gravitational force from both the Earth and the Moon.

X should be marked where the graph is at a maximum since $-\frac{d\phi}{dr} = g = 0$

3(b)(ii)

GPE gained = (potential at neutral point – potential at earth's surface) m

$$6.1 \times 10^{10} = [\phi_0 - (-62.3 \times 10^6)]1000$$

where ϕ_0 = potential at neutral point

$$\phi_0 = -1.3 \times 10^6 \text{ J kg}^{-1}$$

3(b)(i)

The satellite is located left of the neutral point,

This corresponds to the net gravitational force on the satellite being directed towards the Earth, allowing the satellite to orbit the Earth.

3(b)(ii)

Let orbital radius be r .

At all times, the satellite will be between the Earth and the Moon.

$$mr\omega^2 = F_E - F_M$$

where m is the mass of the satellite

$$\frac{GM_E m}{r^2} - \frac{GM_m m}{(D - r)^2} = mr\omega^2$$

O4(a)(i)